

Software Design Document

The Water Management Log

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1. Introduction

1.1. Product Scope

This C++ console-based water-management application allows the user to track their water consumption levels based on common household activities like laundry, shower time, etc., providing not only basic usage statistics but also recommendations for water management. Furthermore, this program uses a vector for the water log's abstract data type implementation and as dynamic storage to allow efficient insertion and deletion of entries.

1.2. Requirements

These are the requirements for this program.

- Must have a water entry class and a water log class (the water log is an ADT)
- There are six files: main.cpp, WaterEntry.h, WaterEntry.cpp, WaterLog.h, WaterLog.cpp. and data.txt
- Each class must have an appropriate mutator, accessor, or constructor method(s)
- Must write the water usage info onto a data log txt file
- Must validate any user integers (most prone to error)
- Must have a menu interface that allows the user to:
 1. Add entry
 2. View entries
 3. Set a personalized water conservation goal
 4. Provide usage statistics
 5. Provide water management tips
 6. Delete entry
 7. Restart log
 8. Exit
- Must read a minimum of 3 water consumption activities
- Must have a minimum of 3 water usage statistics and 3 water conservation tips
- The log must keep track of specified months (April activity goes under April, June goes under June, etc.) AND the entries for each month must be in chronological order
- Must have a minimum of four test cases
- The formulas are as follows:
 1. Shower = Minutes * 2 (this is GPM)
 2. Laundry = loads * 40 (this is per cycle)
 - Minutes for laundry = loads * 45
 3. Dishwashing = loads * 6 (this is per cycle)
 - Minutes for dishwashing = loads * 65
 4. Faucet = Minutes * 2.2 (GPM)
 5. Cost = (Total Gallons in the month * 0.01 ← this is price per gallon) + 30 (fixed fee)
 6. Goal gallons = people * 300

1.3. Assumptions

These are the assumptions made in the design and implementation of the program. It also covers the program's execution.

- The user should use the menu options logically (i.e., adding an entry THEN deleting an entry)
- The user should only enter valid menu options; although data is validated for ints, it is not for strings. However, this should not cause a program to crash
- The user can specify how many activities they want to keep track of in their log
- For the sake of generality, when calculating specific values, like how much water is used, a standard formula is used. For example, we might use a formula calculating how much water a shower uses for x number of minutes rather than calculating based on what shower model it is, or if the shower was turned on/off, etc.
- The user should not add multiple entries for the same date
- The user will be using this application daily (for the sake of the goal functionality)
- When there is no goal, the goal will say “N/A” on the log
- The goal will use a standard of 30 days in a month
- If the user washes dishes by hand, they can just enter it under faucet minutes
- The average American household uses 10k gallons monthly
- The total usage statistics (not the monthly) will not track a goal when outputting the statistics
- An Olympic-sized swimming pool is 660k gallons
- The “restart log” method also restarts the goal to “N/A.”

2. Design

2.1. Class Diagram

WaterEntry
-string date -int month -int showerMinutes -int laundryLoads -int dishwasherCycles -int faucetMinutes
+WaterEntry(date, month, showerMins, laundryLoads, dishwasherCycles, faucetMinutes) +double calculateTotal() +double getTotal() +int getMonth() +string getDate() +int getShowerMinutes() +int getLaundryLoads() +int getDishwasherCycles() +int getFaucetMinutes()

WaterLog (ADT)
-vector entries -string goal -bool entryExists(string date) -int getValidInt(string prompt) (VALIDATES USER DATA)
+WaterLog() +void addEntry() +void deleteEntry() +void displayEntries() +void saveToFile(string filename) +void loadFromFile(string filename) +void printUsageStats() +void makeGoal() +void displayTips() +void clearLog()

****WaterLog contains a vector of WaterEntry objects**

2.2. Pseudo Code

(This pseudo code emulates the main function in main.cpp to demonstrate program functionality so that no function definitions will be implemented)

Welcome the user to the water log system.

Load our data from our text file from the start.

While the user is not done using the program (menu option 8):

 Ask the user for a menu option.

 If the menu option is 1:

 Make a call to the AddEntry() method
 Ask the user for the date and all necessary information regarding usage
 Create a water entry object
 Add to the log and save to the file
 Tell the user “Done! Your data entry has been added to the log.”

 If the menu option is 2:

 Make a call to the displayEntries() method to show all of the entries for the log so far by each month, and it specifies the year at the top
 **A note for display entries: if the user is not meeting their goal, the water log will make mention of that.

 If the menu option is 3:

 Make a call to the makeGoal() method
 Ask for the number of people in the house
 Output a recommended goal to stay under (Ex, “Based on your inputted data, the number of gallons you should not exceed a month (from U.S. EPA) is X# of Gallons.”
 Save the goal to the file

 If the menu option is 4:

 If the user wants to see the statistics by month:
 Make a call to the printUsageStats() (see more about it in the next line)
 Ask the user for the month (we then filter entries by matching the month field in the WaterLog’s vector)

 Display stat 1 (most used activity for the month by comparing the sums of each activity), then ask the user to enter “C” to continue to stat 2 (total gallons used for the month and if they are on track by

comparing) stat 3, (how much money that typically runs), and stat 4 (relate their monthly statistics to another arbitrary statistic)

Else (this is for the entire water log):

Make a call to the printUsageStats() (see more about it in the next line)

Display fact 1 (most used activity based on the entire log), then ask the user to enter "C" to continue to fact 2 (total gallons used for the whole log) and 3 (relate their total statistics to another arbitrary statistic).

If the menu option is 5:

Make a call to the displayTips()

If the menu option is 6:

Ask the user for the date

If found:

Make a call to the deleteEntry() (asks user for the date)

Delete the entry and save it to the file

Tell the user, "The data entry for [date] has been removed from the log."

Else:

Tell the user, "Sorry, no entry was found."

If the menu option is 7:

Ask the user if they are sure they want to clear the log [Yes/No]

If yes:

Make a call to the clearLog()

Save the log to the file

Tell the user "Done. The log has been cleared."

If no:

Tell the user "Got it."

Break and display the menu options again.

If the menu option is 8:

Break from the loop

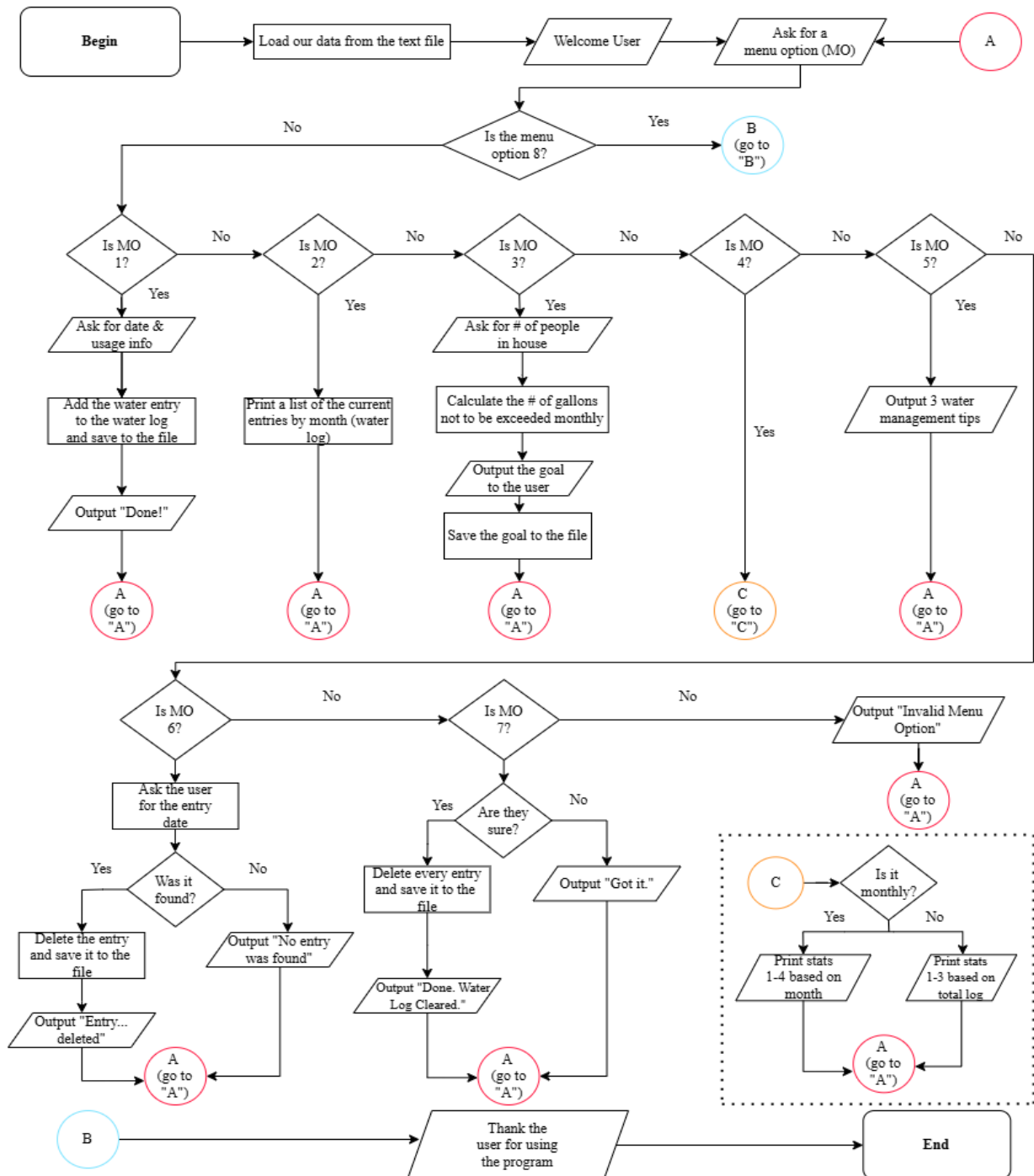
If the menu option is outside the given range:

Print "Invalid menu option."

Restart the loop and ask for another menu option.

Thank the user for using the program and close the file.

2.3. Flow Charts



****THE TEXT FILE WILL ONLY CONTAIN “GOAL:” AND SPACE-SEPERATED ENTRY FIELDS**

3. User Interface

This program will be run on a command line prompt. When the program starts, it displays a welcome message and a menu with eight menu options that the user can perform (listed below for reference). The user should use the menu options logically and carefully (such as adding an entry to the vector and THEN removing it). For any menu option affecting the log, the data will be saved to a text file, ensuring data is persistent across each program execution.

The user may utilize the menu options as many times as they wish. Once the user enters menu option 8, the program will display a message thanking the user for using the program and thus ending it. The user may run the program as many times as they wish.

Menu Options:

1. Add entry
2. View entries
3. Set a personalized water conservation goal
4. Provide usage statistics
5. Provide water management tips
6. Delete entry
7. Restart log
8. Exit

4. Testing

I will test if the program is correct and ensure that all the menu options function as intended using the following test cases. The log records everything in minutes.

For our purposes, some menu options may be used with other menu options in a given test case to demonstrate their functionality.

For brevity, I will not include prompts asking the user for water entry details in the expected output, nor the output of a menu option used to demonstrate the functionality of a different menu option (unless it is needed for that demonstration).

I will display the welcome and closing messages only in the first and last test cases, respectively.

We will have 3 dates preloaded below for the sake of testing (and may be included in the final submission for 3B); by extension, they do not follow the rule that they must be used consecutively.

Note: The spacing may not be consistent in this document, but will be later implemented to have that “table” consistency. Also, the field entries in the text file will not exactly match the order of the field entries in the test case (there is an extra one in the file)

4/9 12 1 1 50

4/11 11 0 1 30

5/11 30 0 1 70

Test Cases are on the following pages.

Test Case 1

Purpose:

This is to test the functionality of the menu options 1 and 2.

Inputs:

1

5/22 10 2 0 2

2

Expected Output:

Welcome to your very own personal water log application! “Waste water today and face a dry tomorrow.”

- Unknown

Water Log

Goal: N/A

----- APRIL -----					
Date	Shower	Laundry	Dishwasher	Faucet	Total (gallons)
4/9	12	45	65	50	180
4/11	11	0	65	30	94
----- MAY -----					
Date	Shower	Laundry	Dishwasher	Faucet	Total (gallons)
5/11	30	0	65	70	220
5/22	10	90	0	70	104

Test Case 2

Purpose:

This is to test the functionality of menu option 3

Inputs:

3
4
2

Expected Output:

Based on your input data, the number of gallons you should not exceed in a month for a 4-person household (from U.S. EPA) is 1200 gallons.

Water Log

Goal: Try to stay under 1200 gallons a month!

----- APRIL -----					
Date	Shower	Laundry	Dishwasher	Faucet	Total (gallons)

4/9	12	45	65	50	180
4/11	11	0	65	30	94
----- MAY -----					
Date	Shower	Laundry	Dishwasher	Faucet	Total (gallons)

5/11	30	0	65	70	220

Test Case 3

(what activity is most used, how many gallons you use, how much that might cost you, and how that represents an environmental impact)

Purpose:

This is to test the functionality of the menu option 4 for the month.

Inputs:

4
Month
4 (this indicates April)
C
C
C

Expected Output:

- The activity that used the most water is from your faucet.
- You have used 274 gallons of water so far in April.
Congratulations, you are on track so far with your goal!
- The total cost for that much water is about \$32.74 a month
- The percent difference between your water household usage and the average American water household usage is approximately a 189% difference.

Test Case 4

Purpose:

This is to test the functionality of menu option 4 for the total.

Inputs:

4
Total
C
C

Expected Output:

- Your most used is from your faucet.
- You have used 494 gallons of water so far.
- That many gallons is about 0.074848% of an Olympic-sized swimming pool!

Test Case 5

Purpose:

This is to test the functionality of the menu option 5

Inputs:

5

Expected Output:

1. Colder water cycles strain the water supply less
2. Installing low-flow shower heads and faucet aerators can help limit water usage
3. Dishwashers are more water-friendly than handwashing
4. Insulating pipes reduces the time waiting for hot water
5. Target shower time should be limited to 5-10 minutes!

Test Case 6

Purpose:

This is to test the functionality of the menu option 6 when a date is found.

Inputs:

6

4/9

2

Expected Output:

The data entry for 4/9 has been removed from the log.

Water Log

Goal: Try to stay under 1200 gallons a month!

----- APRIL -----					
Date	Shower	Laundry	Dishwasher	Faucet	Total (gallons)

4/11	11	0	65	30	94

----- MAY -----					
Date	Shower	Laundry	Dishwasher	Faucet	Total (gallons)

5/11	30	45	65	70	220

Test Case 7

Purpose:

This is to test the functionality of the menu option 6 when no date is found.

Inputs:

6
9/22
2

Expected Output:

Sorry, no entry was found.

Test Case 8

Purpose:

This is to test the functionality of menu option 7 when the user clears the log. Note: I decided to omit the test case for the user not wanting to clear the log because it is self-explanatory (it restarts the menu system). The expected output is also subject to change.

Inputs:

7
Yes
2

Expected Output:

Done. The log has been cleared

Water Log

Goal: N/A

Test Case 9

Purpose:

This is to test the functionality of the menu option 8 and ensures data is persistent.

Inputs:

8
2

Expected Output:

Welcome to your very own personal water log application!

“Waste water today and face a dry tomorrow.”

- Unknown

Thank you for using the water log system. Hope to see you tomorrow!

Water Log

Goal: N/A

----- APRIL -----
Date Shower Laundry Dishwasher Faucet Total (gallons)

4/9 12 45 65 50 180

4/11 11 0 65 30 94

----- MAY -----
Date Shower Laundry Dishwasher Faucet Total (gallons)

5/11 30 45 65 70 220